1m Lidar Target Stand Instructions

Tools

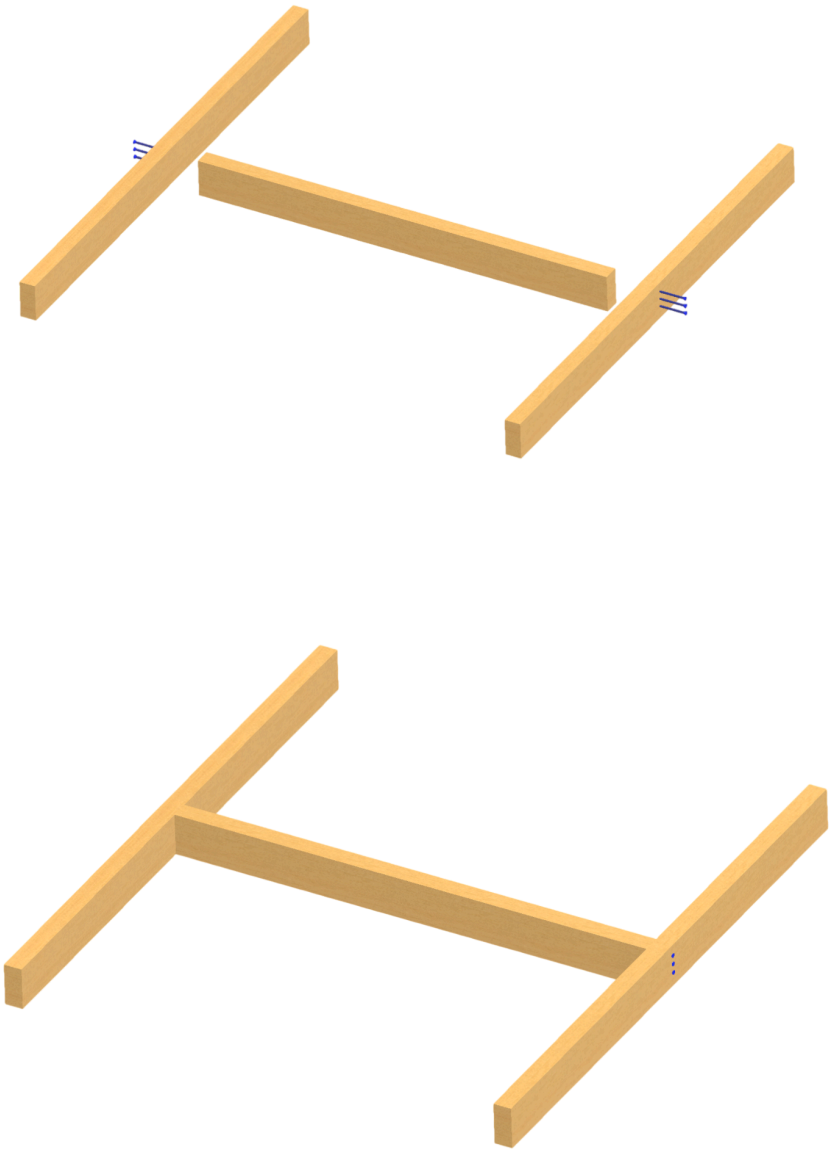
- Cordless Drill
- 5/16-inch Twist Bit
- Chop saw (to cut down lumber)
- 3D Printer

Materials

Item	**Quantity**
2x4 lumber, 40-inch length	1
2x4 lumber, 48-inch length	2
2x6 lumber, 8-inch length	2
4-inch x ⅞-inch x 4-inch Corner L-brace	4
2 ½ inch Deck Screws	14
1-inch Wood Screws	8
5/16-inch, 3 ½-inch long Hex Bolts	4
5/16 inch Washer	32
5/16 inch Wing Nut	4
7-foot signpost	2
3D-printed target bracket for corner support	4
5/16-inch, 2-inch-long Hex Bolt	12
5/16-inch Hex Nut	12

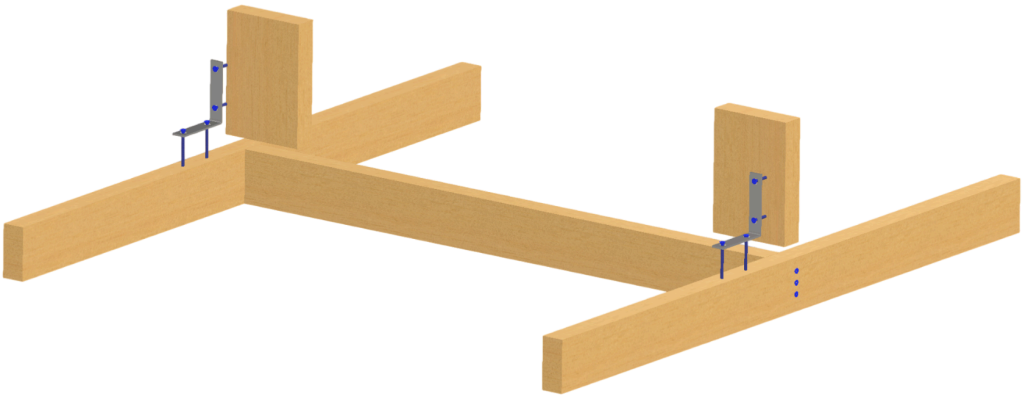
Step 1

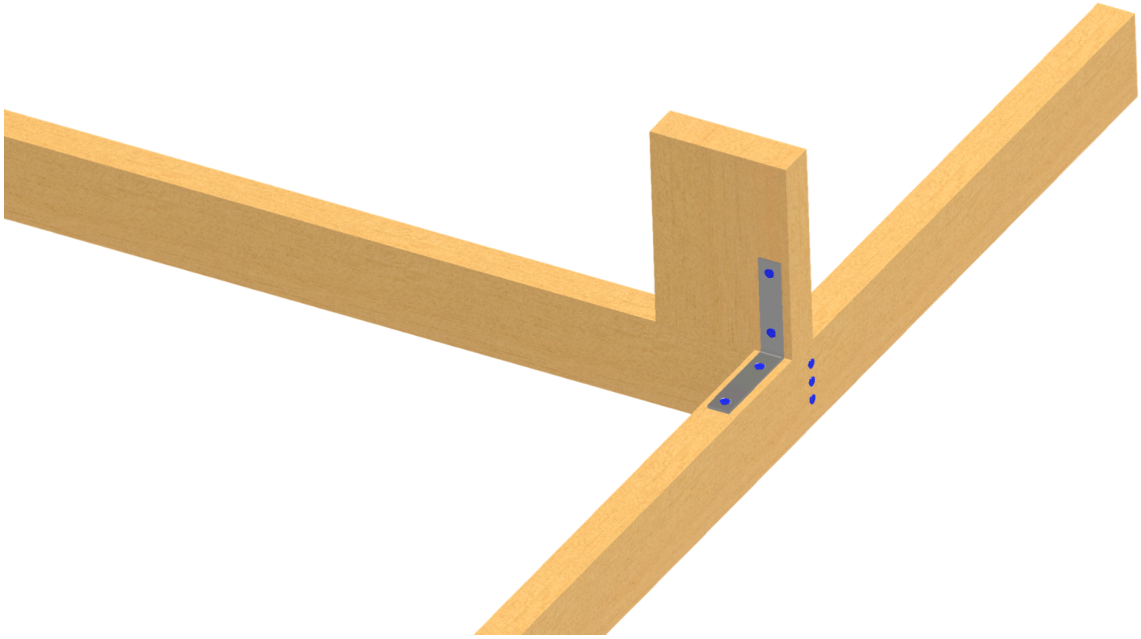
Align the 40-inch board in the center between the 48-inch boards to form an “H” shape, ensuring the middle board is 23.25 inches from the end of each 48-inch board. Secure with 3 2.5-inch deck screws driven through the 48-inch boards into the end of the 40-inch board:



Step 2

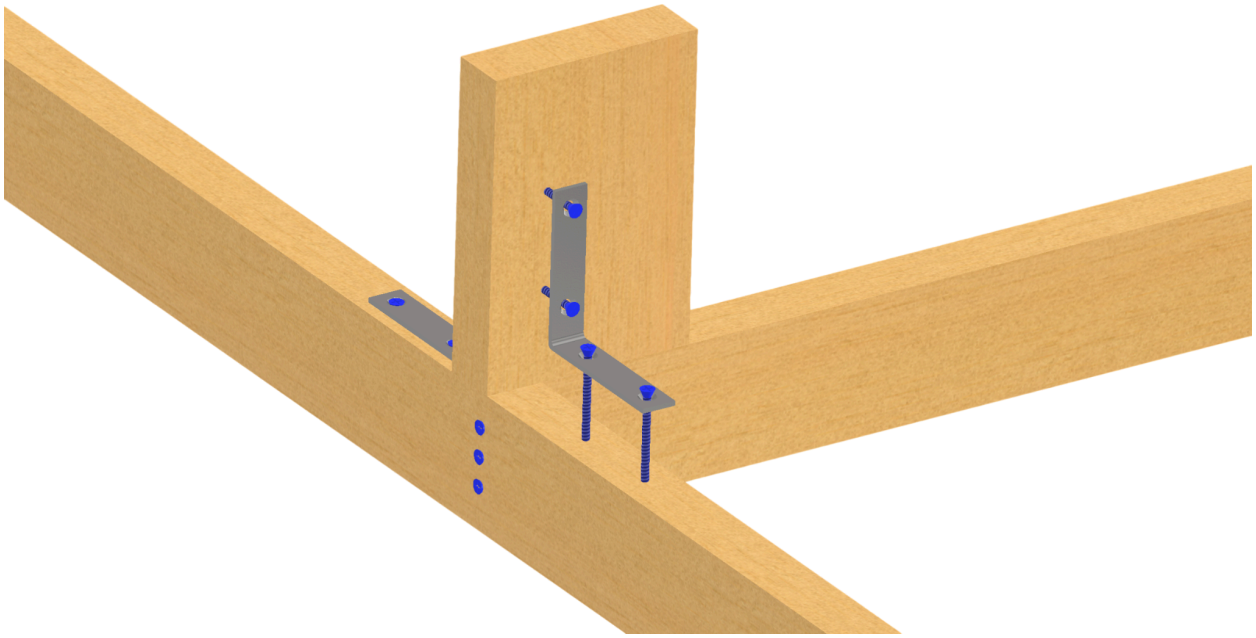
Place each 2x6 board on top of the “H”-shaped base, aligning each 2x6’s width parallel with the 40-inch board and its end with the 48-inch boards. Place two 4-inch corner brackets against the face of the 2x6 boards and along the 48-inch support base. Use 2 ½-inch deck screws to screw into the 48-inch board, and 1-inch screws to screw into the face of the 2x6 board:

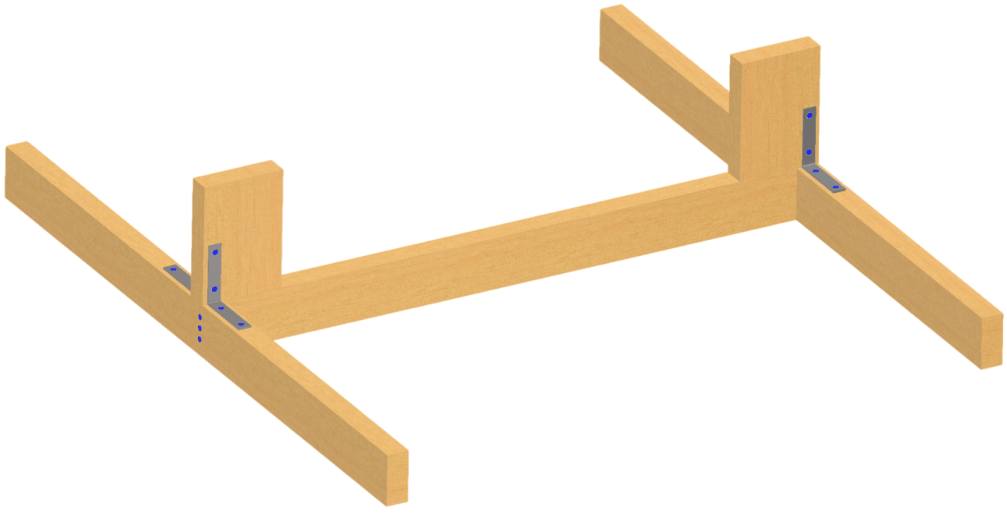




Step 3

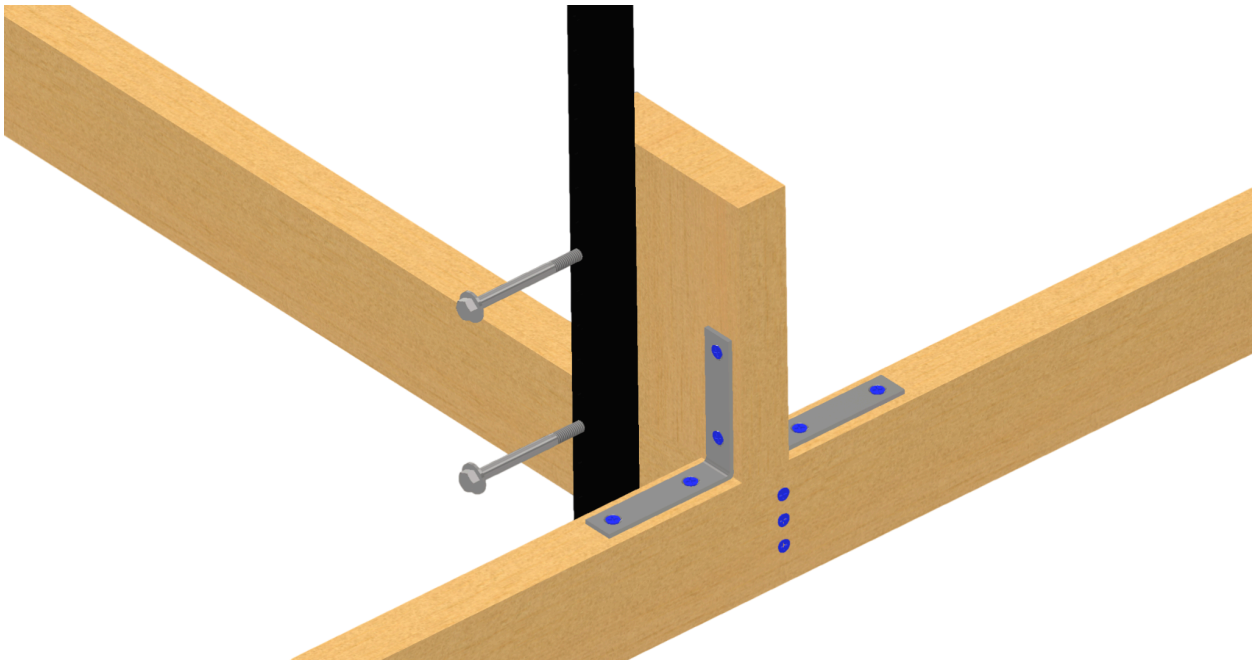
Repeat Step 2 on the other side so that there is an L-corner bracket on each side of the 2x6 upright supports. This creates four L-corner brackets supporting the upright supports.

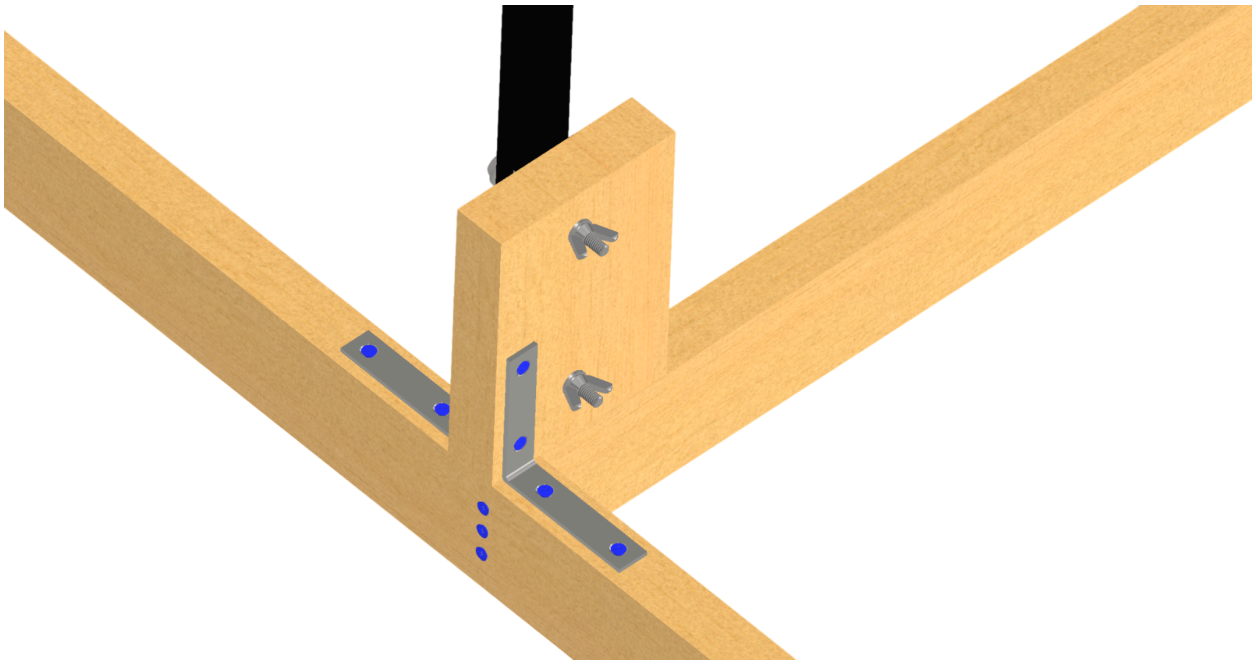




Step 4

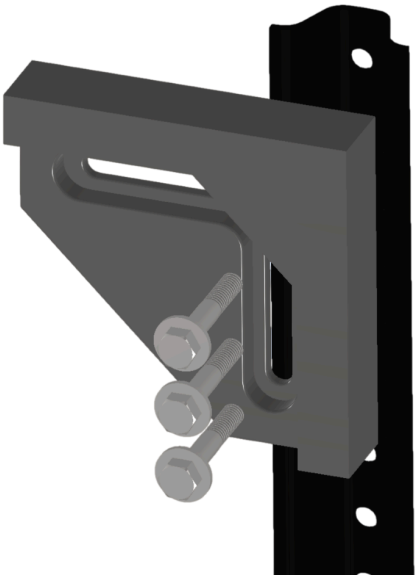
Align the signpost against the 2x6 board at the bottom brace corner. Drill one hole near the bottom of the 2x6 and another near the top to bolt the post to the base. Then install 3 ½-inch bolts through the post and the wooden base, using a washer on each side, and tighten against the base with a wing nut. Repeat for the other side so that both signposts are held upright against the base:

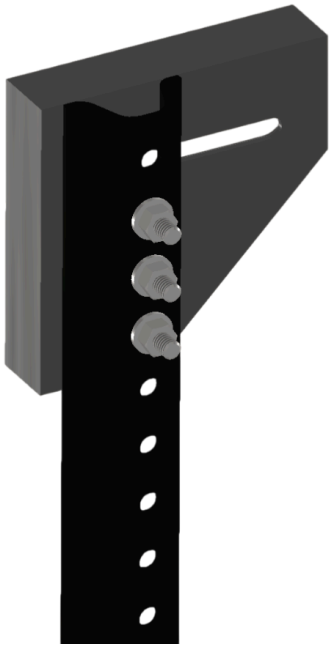




Step 5

At the top of each signpost, mount the 3D-printed corner brackets so the brackets face toward the center of the signpost. Then, align the lower bracket so there is a 1 m gap between the two trays that will support the target. Mount each bracket with three 2-inch bolts, with a washer on each side of the post, and tighten with a hex nut.









Step 6

Install the 1 m target into the brackets by loosening the bolts on one signpost base, tilting the post outward, installing the target, then tilting the post back into place to hold the target's corners. Reinstall the bolts and ensure the target is snug against all corners.

